Orbit deficits on a contiguous interval

Human Compression project: L044 continuation

Let $G = \langle \rho_0, \dots, \rho_{r-1} \rangle$ act transitively on a finite set Ω of size n, where the generators are involutions and $[\rho_j, \rho_k] = 1$ for $|j-k| > 1$. An edge of colour j is *ambient-eligible* if its endpoints lie in different orbits of $G_j = \langle \rho_k : k \neq j \rangle$. Fix a nonempty contiguous interval $I = \{a, \dots, b\}$, set $H = \langle \rho_j : j \in I \rangle$ and $U = \mathrm{supp}(H)$, and suppose that every colour in I has at least two ambient-eligible edges. Select exactly two per colour on vertex set U.

Theorem 1. *In a square-only selection minimizing the number of squares, every proper nontrivial H-orbit has a tree component. If H is intransitive and has c nontrivial orbits and $o(H)$ total orbits on Ω, then*

$$2|I| \leq |U| - c = n - o(H). \tag{1}$$

No intersection property, parity assumption or primitivity is needed.

Proof. We use the complete accepted proof of the ambient terminal-orbit deficit, including its paid paired-cut and equality-localisation proofs. Here is the exact adaptation; none of those arguments is being applied with local H_j eligibility.

The orbit-cut exchanges and square-block lemma remain valid for ambient-eligible edges of any selected colours. The square-only normal form and its minimum-square property use ordinary slides and the global middle move. When a square has colours $i-1, i+1$ in I, contiguity gives $a < i < b$, so $\rho_i \in H$. Thus these moves preserve actual H-orbits. In particular every minimum-square selection has at most one square per component, just as in the terminal proof.

Fix a proper nontrivial H-orbit O and suppose it has no tree component. Its selected edge count is then $E_O = |O|$. The localisation lemma, the ordinary-slide partition argument and the middle-transport argument from the accepted proof are unchanged: all square-colour eligible edges lie in the square's component; ordinary slides preserve its vertex set; and middle moves transport the component partition in O while preserving $E_O = |O|$. These statements used only $\rho_i \in H$, not that I was terminal. In particular every full-graph neighbour of a square vertex stays in that selected component.

Full transitivity supplies a full edge leaving O. Minimize, over all minimum-square selections having $E_O = |O|$, a connector consisting of a selected path from a square to the inside endpoint of such an edge, followed by that single exit edge. Its exit colour x is outside I, because each generator in I preserves O. The selected path is nonempty by the neighbour assertion. An ordinary first-edge slide shortens the connector by the accepted localisation argument, including its exclusion of a second occurrence of the first colour farther down the selected path.

In the remaining case the first edge has colour i and the square colours are $i-1, i+1$. Since $a < i < b$ and $x \notin [a, b]$, we have $|i-x| \geq 2$. Hence ρ_i commutes with the exit generator on *either* side of the interval. Retain the square and transport all other selected edges by ρ_i. Transport the connector after its first edge as well, including its full exit edge. This is an actual exit edge of colour x; its outside endpoint remains outside O, since $\rho_i \in H$. Its selected path now begins on the retained square and is shorter by one. Any coincidence can be removed from the walk to shorten it further. This is the same contradiction as in the terminal proof. Therefore O has a tree component.

For intransitive H, every nontrivial orbit is proper. Each supplies at least one unit of vertices minus edges; square components supply zero. Summing gives $2|I| \le |U| - c$. The fixed points contribute $n - |U|$ further H-orbits, giving the final equality in (1). □

Corollary 2 (A middle interval with two outside budgets)**.** *Suppose* $0 \le h < g \le r-1$, $g \ge h+2$, *and every colour in* $I = \{h+1, \dots, g-1\}$ *has two ambient-eligible edges. Let* $H = G_I$ *be intransitive, with support* V *and* c *nontrivial orbits. Suppose sets* $X, Y \subseteq \Omega$ *satisfy*

$$V \subseteq X \cap Y, \quad X \cup Y = \Omega, \quad 2h \le n - |X| - 1, \quad 2(r-1-g) \le n - |Y| - 1.$$

Then $2r \le n + 2 - c$. *In particular* $c \ge 3$ *implies* $r \le \lfloor (n-1)/2 \rfloor$. *If* $c = 2$, *the only failure of that target has even* n *and* $2r = n$, *and forces equality in both outside budgets,* $V = X \cap Y$, *and exactly one tree component in each nontrivial* H*-orbit of a minimum selection.*

Proof. Theorem 1 and the two budgets give

$$\begin{aligned} 2r &= 2h + 2(g-h-1) + 2(r-1-g) + 4 \\ &\le (n - |X| - 1) + (|V| - c) + (n - |Y| - 1) + 4 \\ &\le n + 2 - c. \end{aligned}$$

For $c = 2$ every inequality must be an equality if the target fails. Equality in the orbit deficit says that the total number of tree components is c, proving the stated necessary conditions. □

Source routing and attribution. This extends the accepted ambient version of the rank step in Cameron–Fernandes–Leemans–Mixer, *Highest rank of a polytope for* A_n, arXiv:1605.09173v1, Proposition 5.27. It is supporting theory for auditing Proposition 5.28, not a recovery of that proposition: the stated support and outside-budget hypotheses of the corollary must still be proved in an application. In particular the reversal of a prefix index is $r - 1 - g$, and the two-orbit equality case is not discarded. Local H_j eligibility is not substituted for ambient eligibility.

The full three-page terminal proof and its accepted paired-cut and equality proofs remain paid dependencies. The earlier global middle-colour move was supplied by the root collaborator and received a distinct review; the two-sided exit adaptation here is the present author's continuation. No census, new formal theorem, whole L044 recovery, original credit or question credit is claimed.